

SMART CLASSROOM ENVIRONMENT MONITORING SYSTEM USING ESP32

Components Required

ESP32
DHT22 Temperature and Humidity Sensor
MQ-2 Gas Sensor
Photoresistor Sensor
OLED SSD1306 Display
Green LED
Red LED
Buzzer
220 Ω Resistor
Jumper Wires
USB Cable
Breadboard

Working

The Smart Classroom Environment Monitoring System uses an ESP32 with a DHT22 sensor, MQ-2 gas sensor, and LDR sensor. The DHT22 measures the classroom's temperature and humidity, the MQ-2 detects smoke or harmful gases, and the LDR measures light intensity. The ESP32 continuously reads data from all the sensors and displays the readings on the OLED display. Under normal conditions, the green LED remains ON. If the classroom becomes dark, the OLED displays "TURN ON LIGHT." If the temperature exceeds the predefined limit, it displays "HOT ROOM." When harmful gas is detected, the red LED turns ON, the green LED turns OFF, the buzzer sounds an alarm, and the OLED displays "GAS ALERT!" This project can be used in schools, colleges, libraries, laboratories, and offices for continuous environmental monitoring and safety.

Connections

Component	ESP32 Pin
DHT22 VCC	3.3V
DHT22 SDA	GPIO 4
DHT22 GND	GND
OLED VCC	3.3V
OLED GND	GND
OLED SCA	GPIO 21
OLED SCL	GPIO 22
MQ-2 VCC	5V
MQ-2 GND	GND
MQ-2 AOUT	GPIO 34
LDR VCC	3.3V
LDR GND	GND
LDR AOUT	GPIO 35
Green LED Anode (A)	GPIO 18 (through 220 Ω Resistor)
Green LED Cathode (C)	GND
Red LED Anode (A)	GPIO 19 (through 220 Ω Resistor)
Red LED Cathode (C)	GND
Buzzer 1:1	GPIO 25
Buzzer 1:2	GND

PROJECT LINK

<https://wokwi.com/projects/470004069877986305>